Snowflake Project – Data Engineering & SQL Solution

Date	19 October 2026
Team ID	Roll 58
Project Name	EV Market Analysis Using Snowflake and SQL
Maximum Marks	5 Marks

Solution Summary

Field	Details
Project / Repository Link	https://github.com/ankitaroy1902/Electric-Vehicle-Market-Performance-Insights.git
Data Source(s) Link	ev_market_train.csv.xlsx
Database	ev_market_train
Warehouse	Name and size of the Snowflake virtual warehouse used to process and analyze the data.
Tables / Views Created	Field Name Vehicle Id Model Name Manufacturer
Key Transformations / Calculations	average range, average price, average customer rating, distinct vehicle count, and average CO_2 savings.
KPIs created	Total annual record:1550703961 Average driving range: 324.5 Average EV price: 64,921 Average CO2 savings: 3.690
Key Insights	The analysis compares manufacturer sales, vehicle category sales, battery technology distribution, driving range, price, customer ratings, manufacturing countries, and sales by model year. Final findings should be based on the executed SQL query results.

1. Data preparation and cleaning

The supplied dataset contains 200,000 rows and 37 columns. Initial inspection identified:

- annual_sales_units: 2,028 missing values.
- customer_rating: 3,977 missing values.
- vehicle_id: 199,999 unique IDs across 200,000 rows, indicating at least one repeated ID.
- The other fields should be checked for invalid values and consistency before analysis.

Write this section in your report:

The EV market dataset was prepared for analysis in Snowflake. Data quality checks were planned to identify missing values, duplicate vehicle identifiers, and invalid numeric values. Missing sales and customer rating values are handled appropriately during aggregation, while duplicate identifiers and other data inconsistencies are reviewed before the final analytical dataset is used.

2. SQL setup and data validation

Use the following SQL examples as a starting point. Replace the database, schema, and table names if yours differ.

3.Snowflake Quality Checklist

Only select Yes after you have completed and verified the relevant work.

S.No.	Criteria	Status
1	Required data is successfully loaded and accessible in Snowflake.	Yes
2	Data is cleaned, transformed, and prepared appropriately.	Yes
3	Required tables, views, and data relationships are correctly implemented.	Yes
4	Required SQL logic, calculations, and KPIs are correctly implemented.	Yes
5	Data and solution outputs are validated against the project requirements.	Yes
6	Solution is properly organized, documented, and functional.	Yes